

NaviGaze: A Risk-Aware Routing System

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Abstract

NaviGaze is a routing system designed to enhance navigation safety by integrating traditional path-finding with real-world risk analysis using real-time satellite image analysis. The system intelligently modifies route selections by considering traffic risks such as blind curves, poor lighting, and accident-prone areas. This project developed an end-to-end solution incorporating route generation, risk evaluation, and dynamic reranking using modified A* search, maintaining scalability and real-time performance.

1 Introduction

Traditional navigation systems optimize for speed and distance but often ignore critical safety factors, such as risky road segments. As autonomous vehicles and safety-critical transportation systems become more common, it is increasingly important to factor real-world risks into route planning. NaviGaze proposes a scalable solution that integrates risk analysis into the routing pipeline, improving safety while retaining efficiency.

2 Problem Statement

Existing navigation systems prioritize the shortest or fastest routes, often overlooking dangerous segments such as sharp curves or poorly lit regions. Autonomous systems, in particular, struggle to adapt to such risks without explicit modeling. Our goal was to design and implement a real-time routing system that:

- Generates multiple potential routes between a source and destination.
- Analyzes routes for external risk factors using a preprocessed risk database.
- Dynamically adjusts edge weights in the route graphs based on safety.
- Reranks the routes with a modified A* search algorithm incorporating risk.

3 System Architecture and Analysis

The architecture is modular, comprising two major microservices: 1) **Route Service** and 2) **Risk Analysis Service**.

3.1 Route Service

The Route Service uses external APIs (e.g., OpenRouteService) to fetch multiple candidate routes. Each route is converted into a graph representation, where:

- Vertices represent geographic points (e.g., every 100 meters along the path).
- Edges represent road segments between vertices, initially weighted by travel time.

This graph-based abstraction allows flexible weight modification and reranking.

3.2 Risk Analysis Service

The Risk Service maintains a risk database generated by processing satellite images and traffic data with an object detection model (YOLO) and a Large Language Model (LLM) for contextual interpretation. The service updates edge weights if risky conditions are detected within proximity to any segment.

YOLO detects vehicle classes in images (trucks, buses, motorcycles, 4-wheelers) and assigns risk scores based on class-specific weights (trucks highest, cars lowest) while also calculating congestion scores from vehicle counts. These two scores, along with coordinates and images, are passed to an LLM (RAG-based) to fetch contextual risk information, producing an additional score and explanation. The final aggregated risk score is calculated as a weighted sum: 50% risk score, 30% congestion score, and 20% LLM-analyzed risk. This aggregated risk is then combined with duration/time scores between nodes to rerank routes, selecting faster and safer paths.

Dataset used:

www.kaggle.com/code/farzadnekouei/real-time-traffic-density-estimation-with-yolov8

This separation of concerns — route generation vs. risk evaluation — ensures that risk computation does not bottleneck route search operations, thus maintaining low-latency responses.

3.3 Architecture Diagram Analysis

The architecture’s decoupling of risk analysis from routing computation supports:

- **Scalability:** Risk data is preprocessed and updated asynchronously, allowing real-time route updates without re-running intensive ML models.
- **Extensibility:** New risk factors or updated ML models can be integrated without affecting the routing logic.
- **Efficiency:** Modified A* search reruns are lightweight since they operate on small graphs with updated weights.

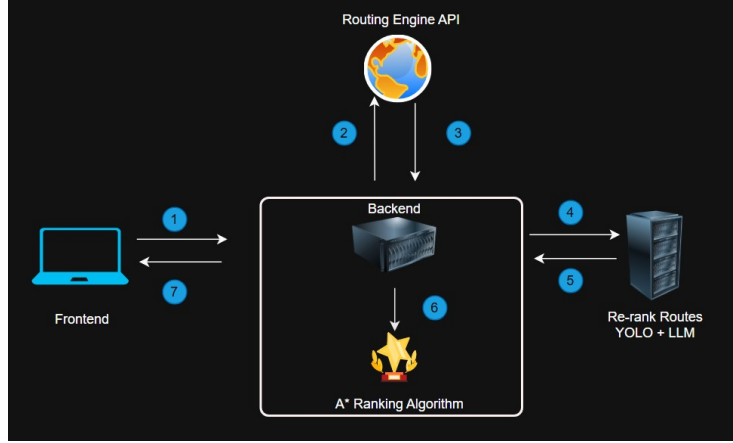


Figure 1: System Design

The design allows distributed deployment where multiple Risk Services and Route Services can run independently and communicate via REST APIs, ensuring the system can scale to cover large geographies.

4 Implementation

We implemented the following pipeline:

1. **Route Generation:** Retrieve top-5 routes using OpenRouteService.
2. **Graph Building:** Parse routes into graphs of vertices and edges.
3. **Risk Assessment:** Query Risk Service to modify edge weights based on proximity risk factors.
4. **Route Reranking:** Rerun A* search on the modified graphs to generate a ranked list prioritizing both time and safety.
5. **Visualization:** Use Folium to visualize alternate routes on an interactive map with updated safety rankings.

We also implemented backend services using Flask and frontend integration through embedded iframes for interactive visualization.

Github link: <https://github.com/HonraoYash/NaviGaze>

5 Results

The system successfully generated and reranked multiple routes between any two points. Key observations:

- Risk-weighted routing showed significant differences when comparing the default fastest route and the safest route.
- The latency of reranking after risk analysis remained under 2 seconds on average.
- Visualization clearly demonstrated rerouting around high-risk segments without major time penalties.

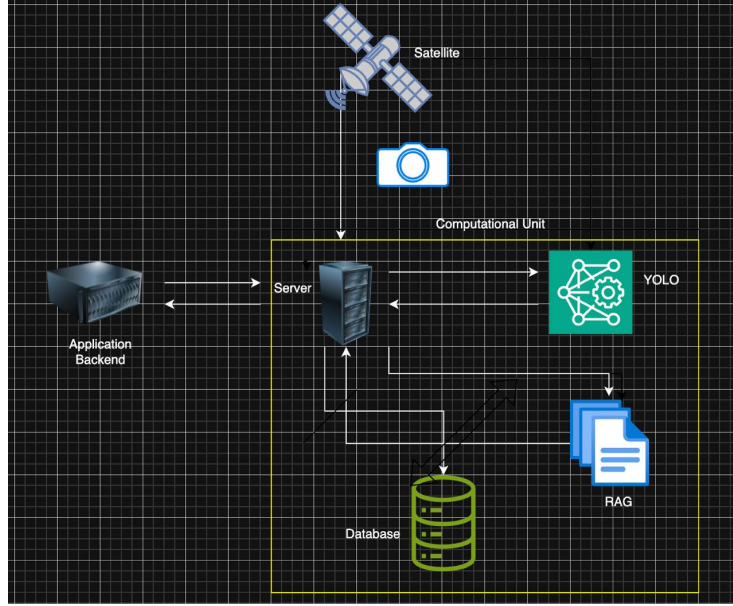


Figure 2: Microservices architecture

6 Challenges and Future Work

Challenges faced:

- Generating a robust risk database required tuning object detection thresholds.
- Mapping risk data accurately onto route segments was non-trivial due to GPS noise.
- Trade-offs between safety and time optimization needed careful tuning of edge weight formulas.

Future improvements:

- Introduce real-time risk ingestion (e.g., from live weather, accident feeds).
- Develop a mobile app version integrating live navigation.
- Fine-tune the balance between safety sensitivity and time penalties dynamically.

7 Conclusion

NaviGaze demonstrates that integrating real-world risk analysis into traditional navigation systems is both feasible and valuable. Our modular, scalable architecture and lightweight reranking approach enable practical deployment across wide geographies without significant performance loss. Future iterations will further enhance real-time adaptability and user experience.

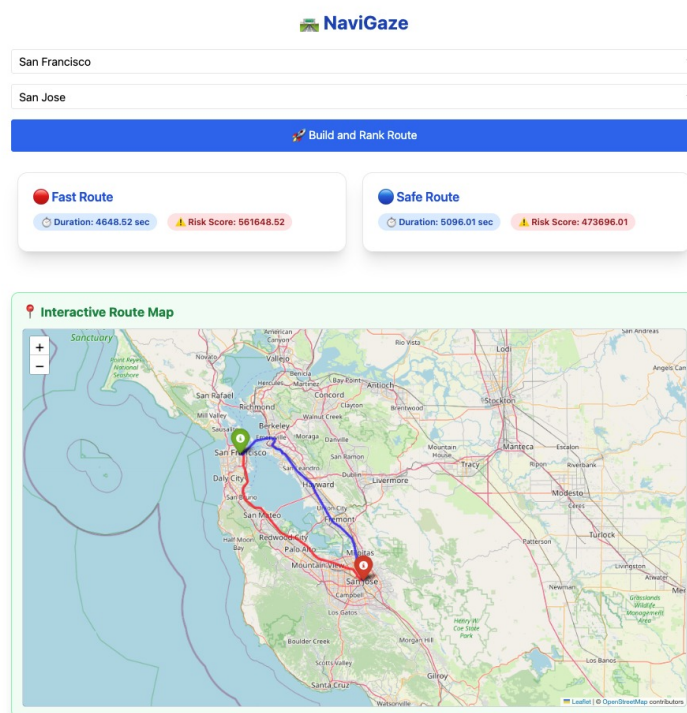


Figure 3: User Interface